MCQ A · transformations, sums and means

After S64; MCQ B after S77.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

1. $E(X) = 3$ and $Y = 2X - 1$. Then $E(Y)$ is:

A. 5 B. 6

C. 2 D. 8

2. $V(X) = 4$ and $Y = -3X + 5$. Then $V(Y)$ is:

A. -12 B. 17

C. 36 D. 41

3. For n independent variables with variance σ^2, the variance of their mean is:

A. $n\sigma^2$ B. σ^2/n

C. σ^2/n^2 D. σ^2

4. The equality $E(X+Y) = E(X) + E(Y)$:

A. requires independence B. requires $X = Y$

C. is false D. holds without independence for finite vars.

5. X is 0 or 2 with probabilities 1/2 each. Its variance is:

A. 0 B. 2

C. 1 D. 4

MCQ B · scope of a guarantee

For each question, only one option is correct. Justify your choice orally before checking the solutions.

6. For $a > 0$, Bienaymé-Chebyshev gives:

A. $P(|X - E(X)| \geq a) \leq V(X)/a^2$ B. $P(|X - E(X)| \geq a) = V(X)/a^2$

C. $P(|X - E(X)| \geq a) \geq V(X)/a^2$ D. $P(X \geq a) \leq E(X)/a^2$

7. For a Bernoulli success frequency, $p = 1/2$, the bound for $P(|F_n - p| \geq 0{,}1)$ is $25/n$. To make it at most 0.05, it is enough to:

A. $n \geq 25$ B. $n \geq 100$

C. $n \geq 250$ D. $n \geq 500$

8. A Chebyshev bound equal to 2 only allows us to deduce:

A. a probability equal to 2 B. the trivial bound 1

C. an impossibility D. a zero probability

9. The law of large numbers states that, for all $a > 0$ fixed:

A. the frequency becomes exactly equal to p B. each new trial reduces the error

C. $P(|F_n - p| \geq a) \to 0$ D. the observations are constant

10. If $X_1 = \cdots = X_n = X$ and $V(X) > 0$, the variance of their mean is:

A. $V(X)$ B. $V(X)/n$

C. 0 D. $nV(X)$

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. A sufficient sample size given by Chebyshev is necessarily the true minimum size.

2. The standard deviation of $aX + b$ is $|a|\sigma(X)$.

3. Independence may be omitted when calculating the variance of a mean.